

RESEARCH METHODOLOGY
BASIC CONCEPT OF RESEARCH



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HANDPRINT IDENTIFICATION WITH CONVOLUTIONAL NEURAL NETWORK

- **Problem Statements**

How can the application of the Convolutional Neural Network (CNN) method and Local Binary Pattern (LBP) feature extraction in palm identification produce high accuracy in the palm identification process?

- **Research Objectives**

To identify palm images using the Convolutional Neural Network (CNN) method with VGG-16 architecture and Local Binary Pattern (LBP) feature extraction, resulting in high accuracy in the palm identification process.

- **Research Type**

This type of research is **Experimental Research**.

Explanation:

1. Experimental research was conducted to test and compare the performance of CNN models with LBP feature extraction in palm identification.
2. The researchers used a palm image dataset, then performed preprocessing, feature extraction, and CNN model training (with VGG-16 architecture).
3. The test results were presented in the form of evaluation metrics such as accuracy, precision, recall, and f1-score.
4. Because this research focused on the application, testing, and evaluation of computational methods in pattern recognition, it falls into the category of experimental research.

RANDOM FOREST CLASSIFICATION OF LABELED RESULTS FROM USER REVIEWS OF THE BRI MOBILE (BRIMO) APPLICATION

- **Problem Statements**

How accurate are the results of random forest from automatic labeling using the vader library and without the library?

- **Research Objectives**

Knowing the results of the accuracy of random forest from automatic labeling using the vader library and without the library.

- **Research Type**

This type of research is **Quantitative Research** with a **Comparative Approach**.

Explanation:

- **Quantitative:** This study measures and compares the accuracy of a classification model (random forest). Accuracy measurements produce numerical values (such as 91.67% and 78.59%), which are the main characteristics of quantitative research.
- **Comparative:** This study compares two different conditions: labeling review data using the Vader library (automatic) and labeling data manually (without a library). The aim is to see the difference in accuracy between the two conditions.

ANALYSIS OF HOAX NEWS SENTIMENT USING NAIVE BAYES

- **Problem Statements**

This study focuses on two main questions: first, what development methods are used to design and create Python desktop applications, and second, whether the implementation of the Naive Bayes algorithm is effective in classifying digital evidence of hoax news sentiment.

- **Research Objectives**

The objectives of this study are threefold: to determine user responses regarding the ease of use of the Python desktop application developed, to analyze the sentiment (positive, negative, and neutral) of the hoax news found, and to determine the accuracy of the classification using the Naive Bayes algorithm.

- **Research Type**

This type of research is **Quantitative Research** with a **Sentiment Approach**.

Explanation:

- **Quantitative:** This study measures and calculates the accuracy, precision, and recall of the Naive Bayes algorithm. The results are numerical values (e.g., 86% accuracy and 88% recall and 88% accuracy), which are the main characteristics of quantitative research. This study also uses the User Acceptance Test (UAT) method with three respondents to measure the percentage of ease of use.
- **Sentiment:** The main focus of the research is to classify digital evidence of hoax news into sentiment categories: positive, negative, and neutral. This is the core of sentiment analysis, a branch of natural language processing that aims to determine the writer's attitude or emotion towards a topic.